

Analog IC Design – Cadence&ADT

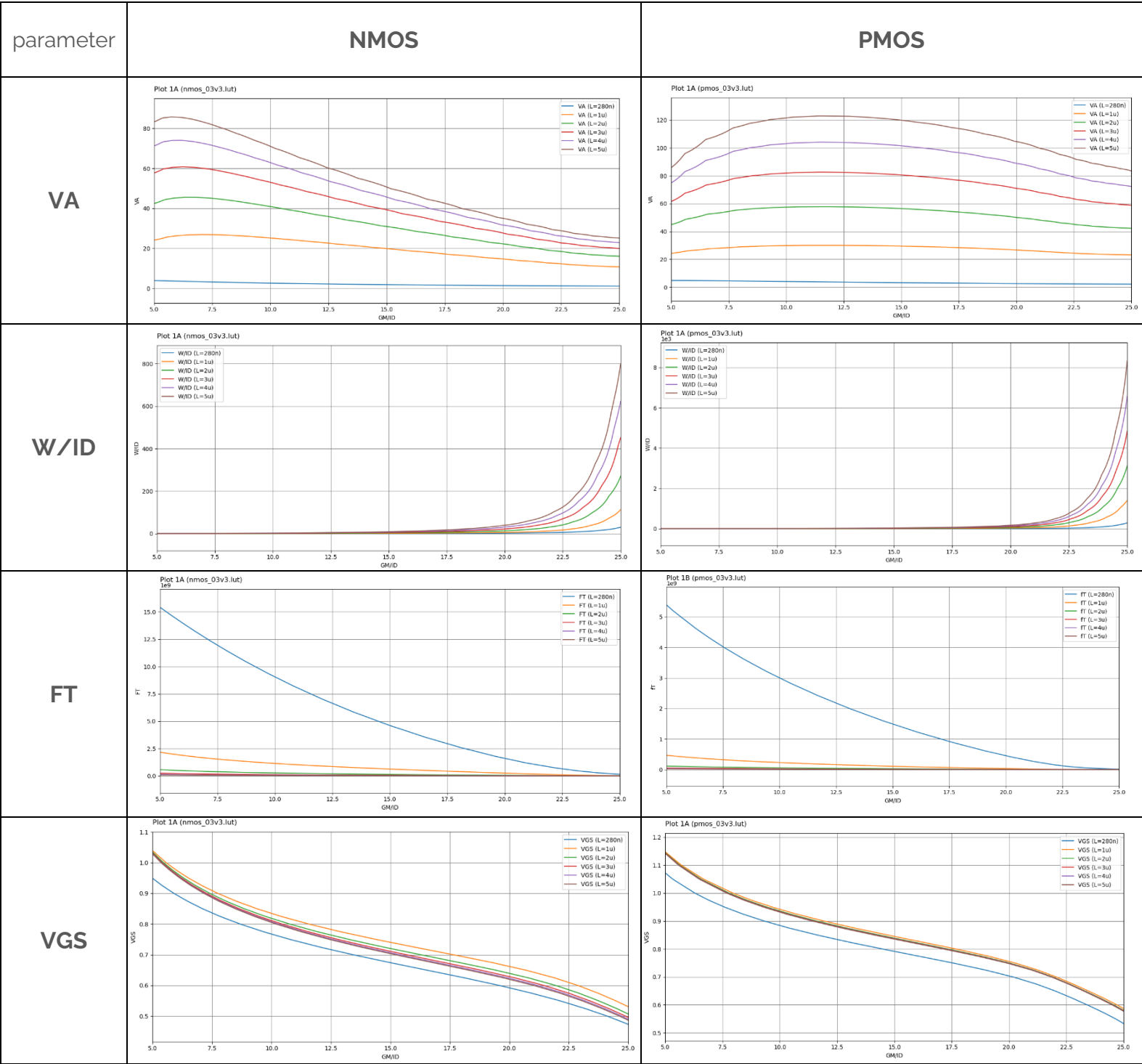
Tools

Fully-Differential Folded Cascode OTA

Lab 11

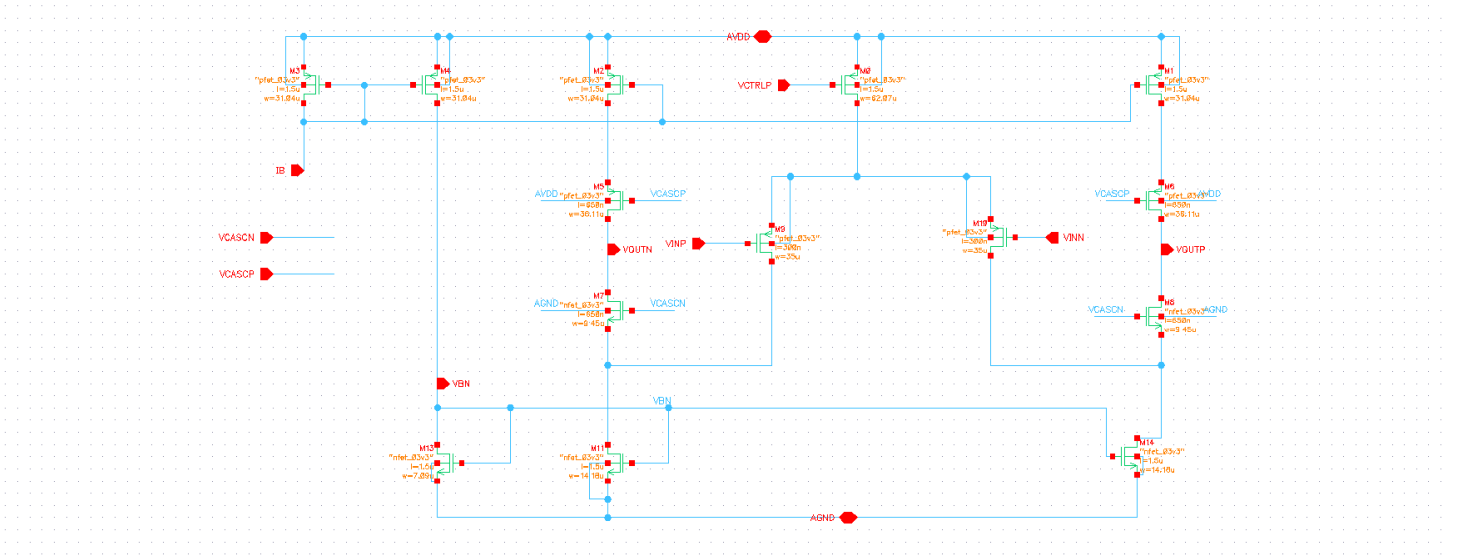
Part 1: gm/ID Design Charts

1. Using ADT Device Xplore, plot the following curves versus gm/ID for both PMOS and NMOS.



Part 2: OTA Design

1. Detailed design procedure and hand analysis.



2. Use the Sizing Assistant (SA) or the design charts in ADT to size the transistors.

$$\beta = \frac{C_f}{C_f + C_{in}} = \frac{1}{3} \gg \gg \gg A_{ol} = 3000$$

CL settling time for 1% error = 100 ns

$$VOUT = VIN \left(1 - e^{-\frac{t}{\tau}} \right) = 0.99 = \left(1 - e^{-\frac{100n}{\tau}} \right) \gg \gg \tau = 21.71 \text{ ns}$$

$$BW_{CL} > \frac{2.2}{2\pi t_{rise}} > 7.33 \text{ MHz} \gg \gg w_u = A_{cl} \times BW_{CL} > 14.66 \text{ MHz}$$

$$gm = 2\pi \times w_u \times C_c > 120 \text{ uS} \gg \gg \text{Assume } gm = 185 \text{ uS}$$

FOR PMOS (Input Pair)

Assume $\frac{g_m}{I_D} = 18.5$ & $L = 300n$ Then compute the input device Transconductance:

$$I_D = \frac{185}{18.5} = 10 \mu A \quad \gg \gg \quad W = 36.32 \mu$$

ID	10u	?	ID	10u	?	ID	10u	?
gm/ID	18.5	?	gm/ID	18.5	?	gm/ID	18.5	?
L	300n	?	L	300n	?	L	300n	?
VDS	1.25	?	VDS	1.25	?	VDS	1.25	?
VSB	0	?	VSB	0	?	VSB	0	?
Stack	1	?	Stack	1	?	Stack	1	?

Results:		Results:		Results:	
Name	TT-27.0	Name	TT-27.0	Name	TT-27.0
1 ID	10u	10 fT	699.6MEG	19 ro	394.3k
2 IG	N/A	11 gm/gds	71.85	20 Ron	125k
3 L	300n	12 VA	3.943	21 VTH	739.6m
4 W	36.32u	13 ID/W	275.3m	22 VDSAT	81.8m
5 VGS	741.4m	14 gm/W	5.017	23 cgg	41.45f
6 VDS	1.25	15 AREA	10.9p	24 cdd	20.38f
7 VSB	0	16 gm	182.2u	25 csg	19.92f
8 gm/ID	18.22	17 gmb	63.43u	26 cdg	14.5f
9 Vstar	109.8m	18 gds	2.536u	27 cgd	4.852f

FOR PMOS (Tail Current Source)

Assume $\frac{g_m}{I_D} = 10$ & $L = 1.5 \mu$

$$I_D = 20 \mu A \quad \gg \gg \quad W = 62.07 \mu$$

Name	TT-27.0	Name	TT-27.0
1 ID	20u	10 fT	102.3MEG
2 IG	N/A	11 gm/gds	353.9
3 L	1.5u	12 VA	35.84
4 W	62.07u	13 ID/W	322.2m
5 VGS	939.6m	14 gm/W	3.182
6 VDS	500m	15 AREA	93.11p
7 VSB	0	16 gm	197.5u
8 gm/ID	9.875	17 gmb	94.81u
9 Vstar	202.5m	18 gds	558.1n

FOR NMOS (Current Source)

Assume $\frac{g_m}{I_D} = 10$ & $L = 1.5\mu$

$$I_D = 20 \mu A$$

>>>

$$W = 14.18\mu$$

	Name	TT-27.0		Name	TT-27.0	
1	ID	20u		10	ft	506.3MEG
2	IG	N/A		11	gm/gds	296
3	L	1.5u		12	VA	29.65
4	W	14.18u		13	ID/W	1.41
5	VGS	825.9m		14	gm/W	14.08
6	VDS	500m		15	AREA	21.27p
7	VSB	0		16	gm	199.7u
8	gm/ID	9.984		17	gmb	77.56u
9	Vstar	200.3m		18	gds	674.6n

FOR PMOS (Current Source)

Assume $\frac{g_m}{I_D} = 10$ & $L = 1.5\mu$

$$I_D = 10 \mu A$$

>>>

$$W = 31.04 \mu$$

	Name	TT-27.0		Name	TT-27.0	
1	ID	10u		9	Vstar	202.5m
2	IG	N/A		10	ft	102.3MEG
3	L	1.5u		11	gm/gds	353.9
4	W	31.04u		12	VA	35.84
5	VGS	939.6m		13	ID/W	322.2m
6	VDS	500m		14	gm/W	3.182
7	VSB	0		15	AREA	46.56p
8	gm/ID	9.876		16	gm	98.76u
9	Vstar	202.5m		17	gmb	47.41u

FOR NMOS (VCASCN)

Assume $\frac{g_m}{I_D} = 15$ & $L = 650n$

$$I_D = 10 \mu A \quad \gg \gg \quad W = 9.45 \mu$$

Name		TT-27.0
1	ID	10u
2	IG	N/A
3	L	650n
4	W	9.45u
5	VGS	912m
6	VDS	750m
7	VSB	500m
8	gm/ID	14.71
9	Vstar	135.9m

Name		TT-27.0
10	ft	1.474G
11	gm/gds	225.7
12	VA	15.34
13	ID/W	1.058
14	gm/W	15.57
15	AREA	6.143p
16	gm	147.1u
17	gmb	43.01u
18	gds	652n

FOR PMOS (VCASCP)

Assume $\frac{g_m}{I_D} = 15$ & $L = 650n$

$$I_D = 10 \mu A \quad \gg \gg \quad W = 36.11 \mu$$

Name		TT-27.0
1	ID	10u
2	IG	N/A
3	L	650n
4	W	36.11u
5	VGS	1.058
6	VDS	750m
7	VSB	500m
8	gm/ID	14.47
9	Vstar	138.2m

Name		TT-27.0
10	ft	299MEG
11	gm/gds	308.9
12	VA	21.35
13	ID/W	276.9m
14	gm/W	4.007
15	AREA	23.47p
16	gm	144.7u
17	gmb	52.32u
18	gds	468.4n

FOR NMOS (Current Mirror)

$$I_D = 10 \mu A \quad \& \quad L = 1.5 \mu \quad \gg \gg \quad W = \frac{10 \times 14.18}{20} = 7.09 \mu$$

$$V_{CASCN} = V_{GSN} + V^* = 0.912 + 0.2003 = 1.1123 V$$

FOR PMOS (Current Mirror)

$$I_D = 10 \mu A \quad \& \quad L = 1.5 \mu \quad \gg \gg \quad W = \frac{10 \times 31.04}{10} = 31.04 \mu$$

$$V_{CASCN} = V_{DD} - |V_{GSP}| - V^* = 2.5 - 0.707 - 0.1507 = 1.2395 V$$

PART 3: Open-Loop OTA Simulation (Behavioral CMFB)

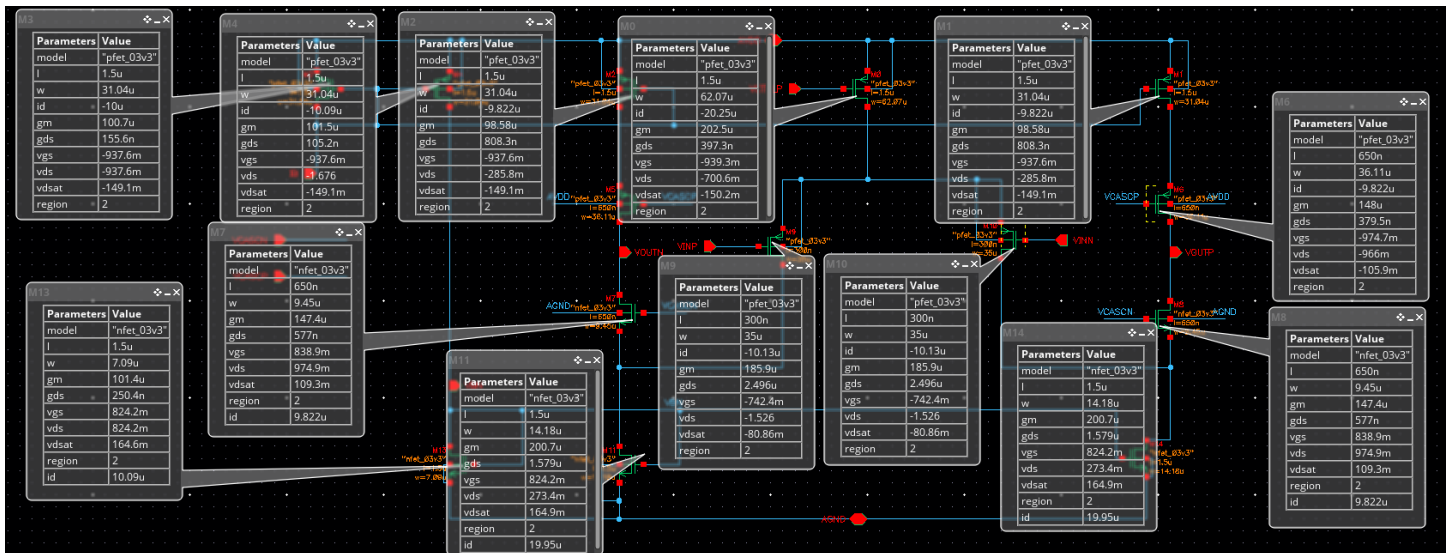
- ❖ Set VICM at the middle of the CMIR

$$VICM = 0.5 V$$

- ❖ Select VREF to maximize the symmetrical output swing

$$V_{REF} = \frac{V_{DD}}{2} = 1.25 V$$

Schematic of the OTA and bias circuit with DC node voltages and transistors OP parameters (id, vgs, vds, vdsat, vth, gm, gds, region) clearly annotated.



What is the CM level at the OTA output?

$V_{out\ cm} = 1.249\ V$. which is approximately equal to V_{REF} which is selected to maximum output swing.

Test	Output	Nominal
FoldedCascode:TB:1	v("/VOCM" ?result "dcOp")	1.249
FoldedCascode:TB:1	vin diff	857.9u
FoldedCascode:TB:1	vout diff	857.9u

What are the differential input and output voltages of the error amplifier? What is the relation between them?

The differential input and output voltages of the error amplifier:

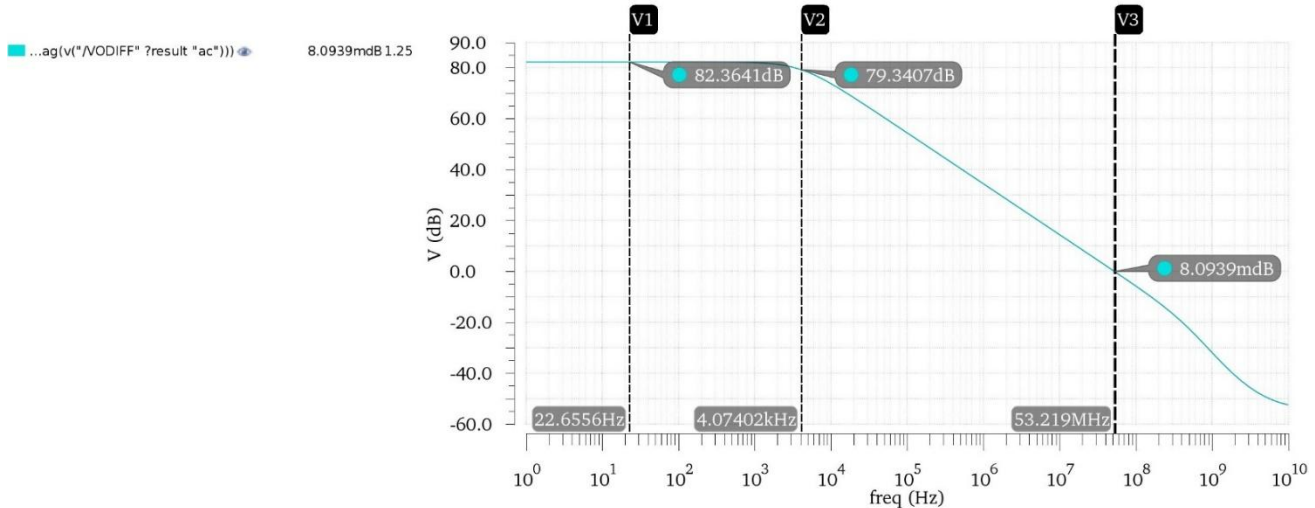
$$\Delta V_{in} = \Delta V_{out} = 857.9\ \mu V$$

The relation between input and output voltage of error amplifier is $\frac{\Delta V_{in}}{\Delta V_{out}} = 1$, error amplifier is ideal and the relation between them is the gain of the error amplifier

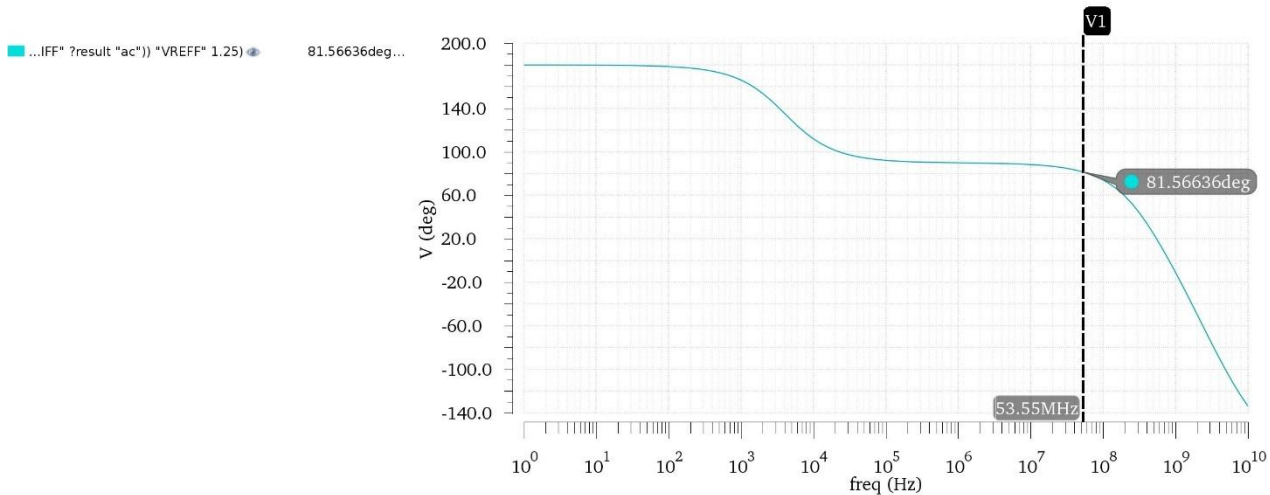
Plot diff gain (magnitude in dB and phase) vs frequency.

dB20(mag(v("/VODIFF" ?result "ac")))

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Calculate circuit parameters (DC gain, BW, GBW, UGF, and PM).

Test	Output	Nominal
FoldedCascode:TB:1	Ao	13.13k
FoldedCascode:TB:1	Ao_dB	82.36
FoldedCascode:TB:1	BW	4.056k
FoldedCascode:TB:1	fu	53.55M
FoldedCascode:TB:1	GBW	53.25M
FoldedCascode:TB:1	phase(v("/VODIFF" ?result "ac...	
FoldedCascode:TB:1	PhaseMargin	81.65

Compare simulation results with hand calculations in a table (use SS parameters from OP simulation in your hand analysis).

$$R_{OUT} = (r_{o8} * (1 + (g_{m8} + g_{mb8}) * (r_{o10} || r_{o4}))) || (r_{o6} * (1 + (g_{m6} + g_{mb6}) * r_{o1}))$$

<i>Parameter</i>	<i>Simulation</i>	<i>Hand Analysis</i>
<i>A_o</i>	13.13k	$g_{m10} \times R_{out} = 13.5k$
<i>A_o_{dB}</i>	82.36	$20\log(A_o) = 82.6dB$
<i>BW(Hz)</i>	4.056k	$\frac{1}{2\pi C_l \times R_{out}} = 4.36k$
<i>F_u</i>	53.55M	$A_o \times BW = 58.58M$
<i>GBW</i>	53.25M	$A_o \times BW = 58.58M$

PART 4: Open-Loop OTA Simulation (Actual CMFB)

FOR PMOS (Input CD)

Assume $\frac{g_m}{I_D} = 15$ & $L = 1.5u$

$$I_D = 5 \mu A \quad \gg \gg \quad W = 50.62u$$

	Name	TT-27.0		Name	TT-27.0
2	IG	N/A	10	fT	50.16MEG
3	L	1.5u	11	gm/gds	1.001k
4	W	50.62u	12	VA	67.82
5	VGS	842.6m	13	ID/W	98.78m
6	VDS	1.25	14	gm/W	1.458
7	VSB	0	15	AREA	75.93p
8	gm/ID	14.76	16	gm	73.82u
9	Vstar	135.5m	17	gmb	35.61u
10	fT	50.16MEG	18	gds	73.72n

FOR NMOS (input Pair)

Assume $\frac{g_m}{I_D} = 15$ & $L = 1.5u$

$$I_D = 5 \mu A \quad \gg \gg \quad W = 11.46 u$$

	Name	TT-27.0		Name	TT-27.0
1	ID	5u	10	fT	276.4MEG
2	IG	N/A	11	gm/gds	403
3	L	1.5u	12	VA	27.22
4	W	11.46u	13	ID/W	436.3m
5	VGS	730.7m	14	gm/W	6.459
6	VDS	700m	15	AREA	17.19p
7	VSB	0	16	gm	74.02u
8	gm/ID	14.8	17	gmb	28.86u
9	Vstar	135.1m	18	gds	183.7n

FOR NMOS (Tail)

$$I_D = 10 \mu A \quad \& \quad L = 1.5 \mu \quad \gg \gg \quad W = \frac{10 \times 14.18}{20} = 7.09 \mu$$

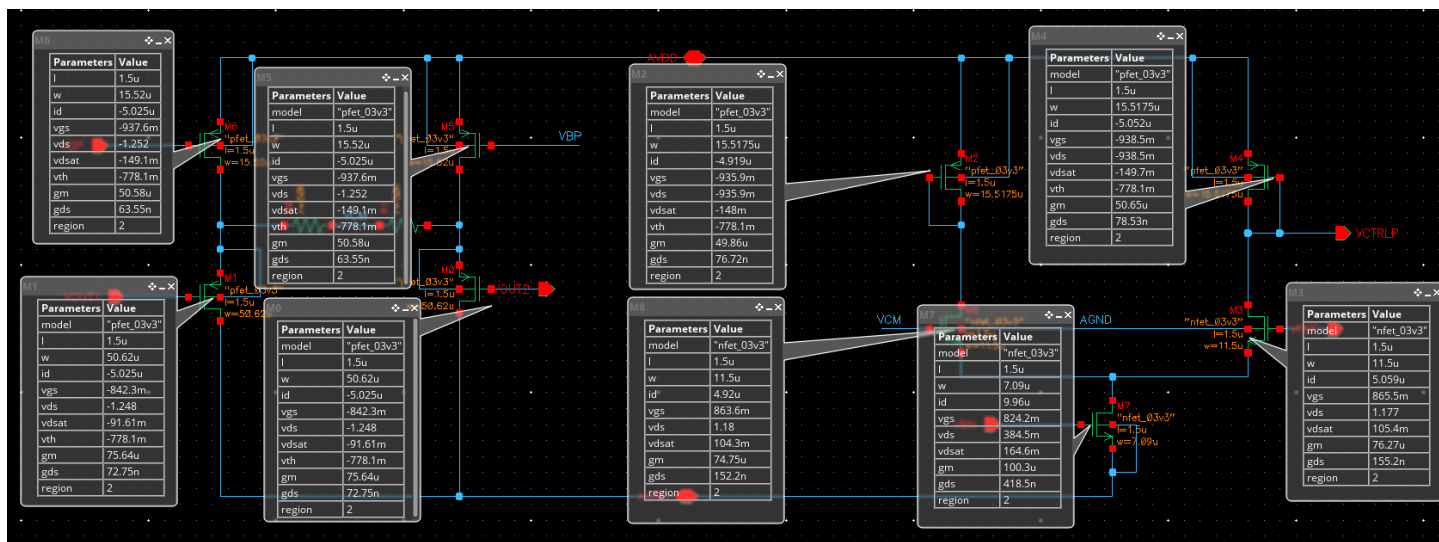
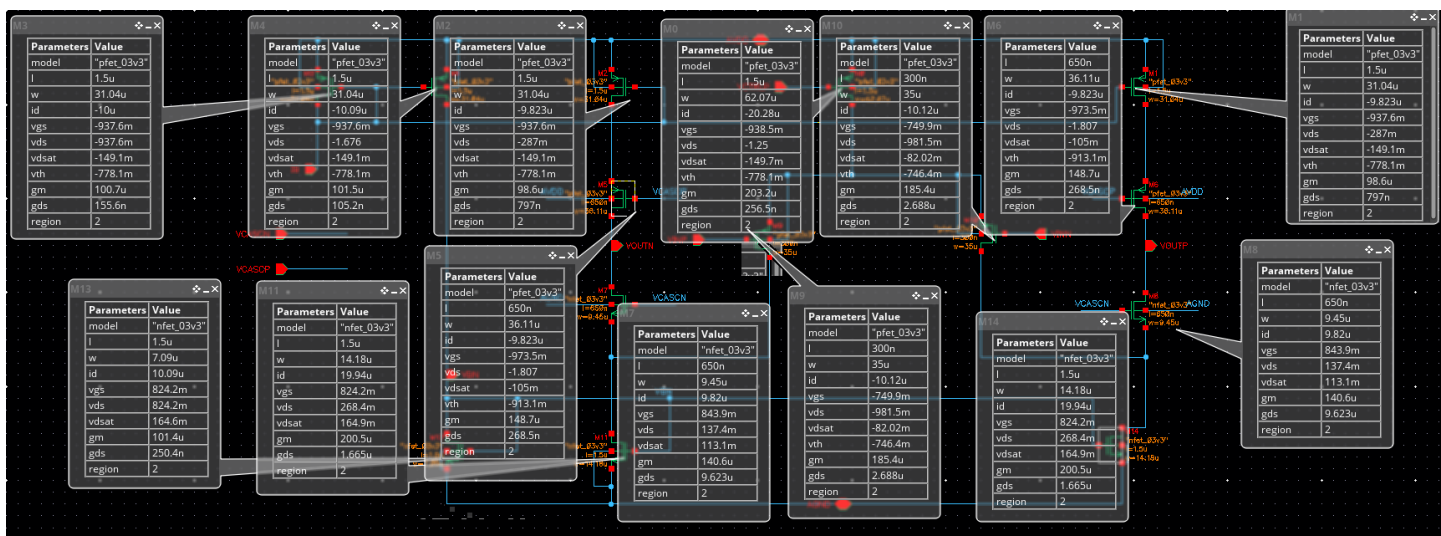
FOR PMOS (Diode Connected)

$$I_D = 5 \mu A \text{ \& } L = 1.5 \mu \gg \gg W = \frac{5 \times 62.07}{20} = 15.5175 \mu$$

FOR PMOS (Load CD)

$$I_D = 5 \mu A \quad \& \quad L = 1.5 \mu \quad \gg \gg \quad W = \frac{5 \times 31.04}{10} = 15.52 \mu$$

Schematic of the OTA and CMFB circuit with DC node voltages and transistors OP parameters (id, vgs, vds, vdsat, vth, gm, gds, region) clearly annotated.



What is the CM level at the OTA output? Why?

$$V_{REF} = \frac{V_{out,low} + V_{out,high}}{2} + V_{GSCD}$$

$$V_{out,low} = V_{Dsat8} + V_{Dsat4} = 0.2783 V$$

$$V_{out,high} = V_{DD} - V_{GSCD} - V^* = 1.5V$$

$$V_{REF} = 1.75V$$

What is the CM level at the OTA output? Why?

Voutcm = 887.6 mV, which is approximately equal to $V_{REF} - V_{GSP}$ (to compensate shift due to common drain) which is selected to get maximum output possible swing.

What are the differential input and output voltages of the error amplifier? What is the relation between them?

- The differential input voltage of the error amplifier:

$$\Delta V_{in} = V_{ref} - V_{ocm} = 200.7 \mu V$$

The differential output voltage of the error amplifier:

$$\Delta V_{out} = V_{BP} - V_{CtrlP} = 835.1 \mu V$$

The differential input and output voltages of error amplifier aren't equal due to small finite open loop gain of error amplifier:

Test	Output	Nominal
FoldedCascode:closed:1	v("/VOCM" ?result "dcOp")	887.6m
FoldedCascode:closed:1	v("/VFBN" ?result "dcOp")	887.6m
FoldedCascode:closed:1	v("/VFBP" ?result "dcOp")	887.6m

Test	Output	Nominal
FoldedCascode:TB:1	v("/I8/VCM" ?result "dcOp")	1.73
FoldedCascode:TB:1	vin diff	200.7u
FoldedCascode:TB:1	vout diff	835.1u

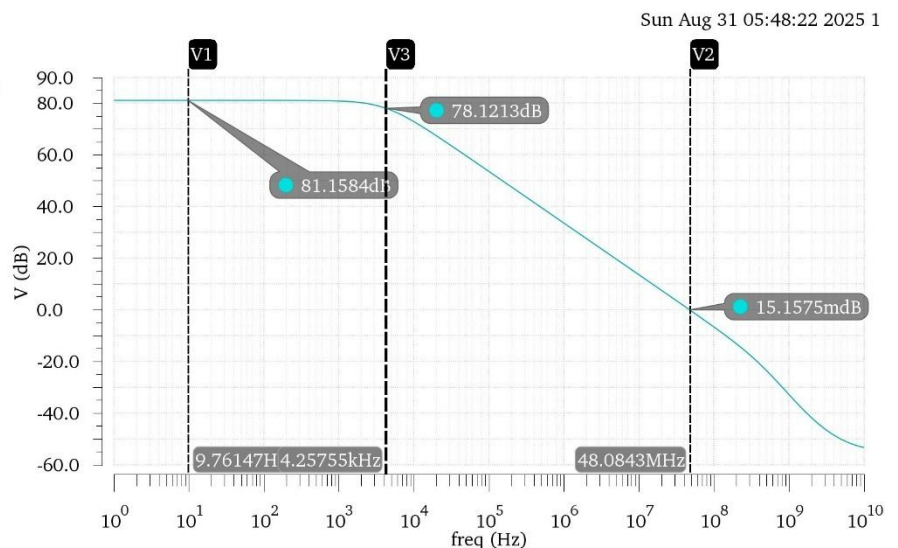
2) Diff small signal ccs:

Plot diff gain (magnitude in dB and phase) vs frequency.

dB20(mag(v("/VODIFF" ?result "ac")))

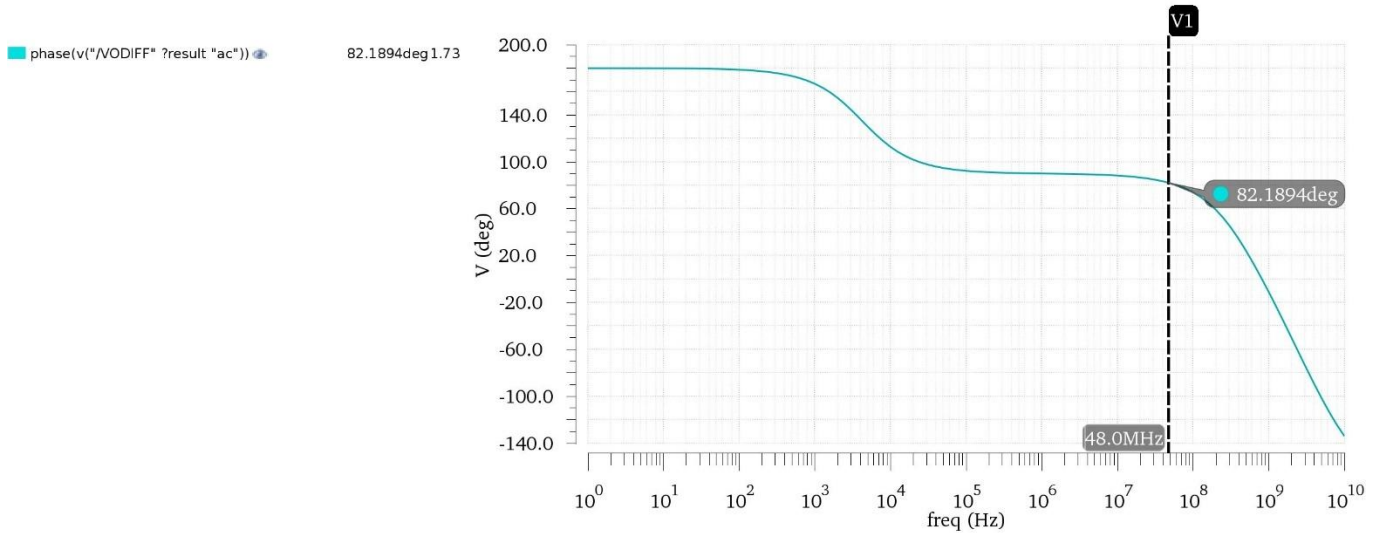
...mag(v("/VODIFF" ?result "ac"))

78.1213dB 1.73



phase(v("/VODIFF" ?result "ac"))

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Calculate circuit parameters (DC gain, BW, GBW, UGF, and PM).

Test	Output	Nominal
FoldedCascode:TB:1	Ao	11.43k
FoldedCascode:TB:1	Ao_dB\	81.16
FoldedCascode:TB:1	BW	4.233k
FoldedCascode:TB:1	Fu	48.34M
FoldedCascode:TB:1	GBW	48.36M
FoldedCascode:TB:1	phaseMargin	82.19

PART 5: Closed Loop Simulation (AC and STB Analysis)

What is the CM level at the OTA output? Why?

$V_{outcm} = 887.6 \text{ mV}$, which is approximately equal to $V_{REF} - V_{GSP}$ (to compensate shift due to common drain) which is selected to get maximum output possible swing.

What is the CM level at the OTA input? Why?

$V_{incm} = 887.6 \text{ mV}$ it is the same value as V_{OCM} as we blocked the dc input voltage when inserted the capacitive FB(C_{in}) it acted as a coupling capacitor, and we shunted CFB by a large resistor to define the DC voltage at the input to be the same as the CM output voltage.

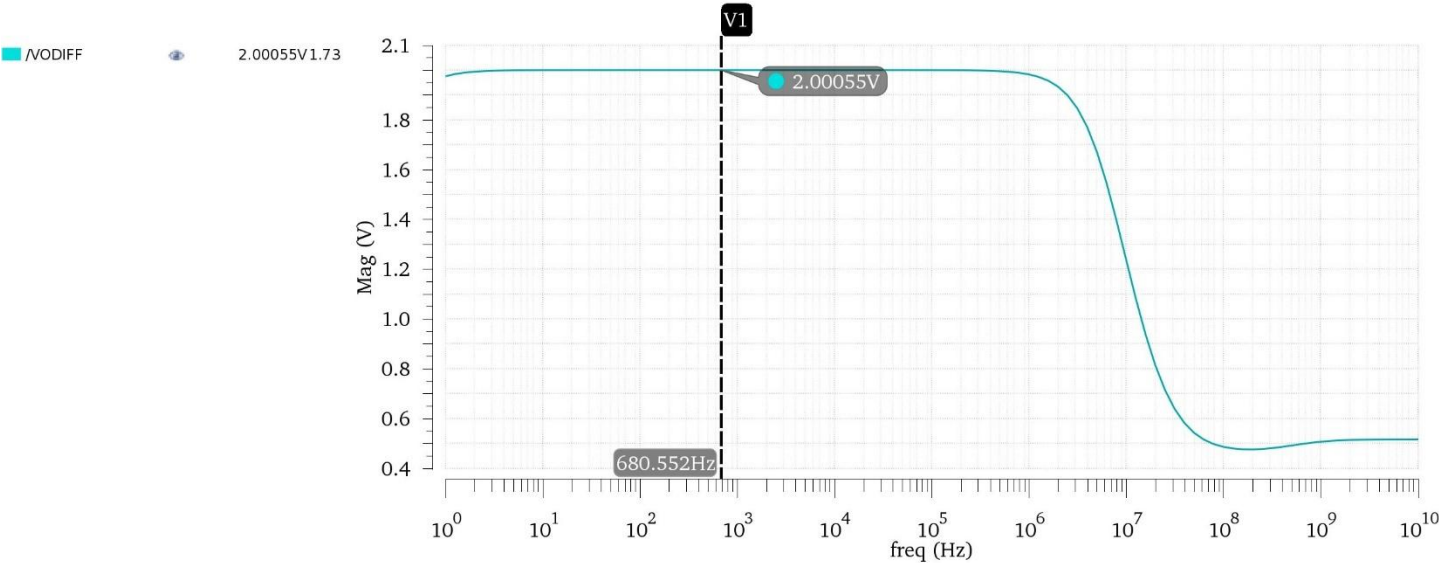
Test	Output	Nominal
FoldedCascode:closed:1	v("/VOCM" ?result "dcOp")	887.6m
FoldedCascode:closed:1	v("/VBN" ?result "dcOp")	887.6m
FoldedCascode:closed:1	v("/VBP" ?result "dcOp")	887.6m

Differential closed-loop response

Plot VODIFF vs frequency

AC Analysis `ac`: freq = (1 Hz -> 10 GHz)

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Use Measures or cursors to calculate circuit parameters (DC gain, CL BW, CL GBW)

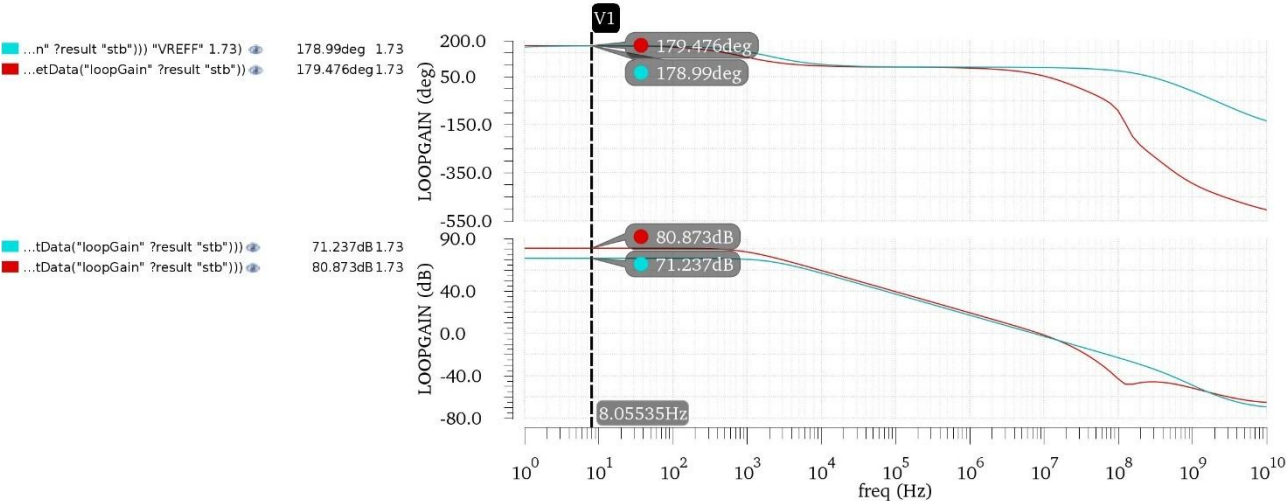
Test	Output	Nominal
FoldedCascode:closed:1	Ao	2.001
FoldedCascode:closed:1	Ao_dB	6.023
FoldedCascode:closed:1	BW	7.956M
FoldedCascode:closed:1	Fu	14.5M
FoldedCascode:closed:1	GBW	15.92M

Differential and CMFB loops stability (STB analysis)

Plot loop gain in dB and phase vs frequency for the two simulations overlaid.

phase(getData("loopGain" ?result "stb"))

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Compare GBW and PM of diff and CM loops. Comment.

Test	Output	Nominal
FoldedCascode:closed:1	Aol	11.06k
FoldedCascode:closed:1	GBWI	9.765M
FoldedCascode:closed:1	BWI	883
FoldedCascode:closed:1	(180 + phaseMargin(getData...	57.39

Figure 2 : CM

Test	Output	Nominal
FoldedCascode:closed:1	Aol	3.687k
FoldedCascode:closed:1	GBWI	7.387M
FoldedCascode:closed:1	BWI	2.004k
FoldedCascode:closed:1	PM	88.83

Figure 1 : DIFF

Differential loop has higher output capacitance than CMFB loop and GBW is inversely proportional with output capacitance so Differential loop will have lower GBW than CMFB loop and therefore phase margin of Differential loop will be higher than phase margin of CMFB loop.

Compare DC LG and GBW of the diff loop with those obtained from open-loop simulation. Comment

Parameter	Diff Loop	Loop Gain
A_o	13.13k	3.687K
$A_{o_{dB}}$	82.36	71.33
GBW	53.25M	7.387M

GBW of the differential closed LG is smaller than GBW of the open LG, DC LG proportional to GBW & DC LG of the closed LG is smaller than DC LG of the open loop, because $LG = A_{ol} * B$ and $B < 1$ due to parasitic capacitance.

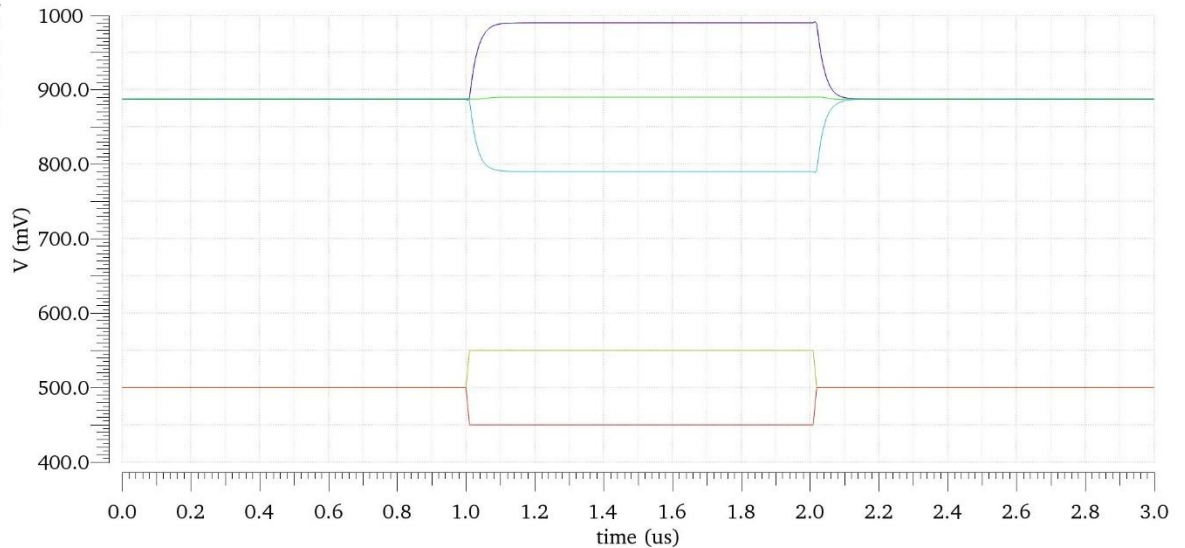
PART 6: Closed Loop Simulation (Transient Analysis)

Differential and CMFB loops stability (transient analysis) + CL settling time: Differential input pulse

Plot the transient signals at VINP, VINN, VOUTP, VOUTN, and VOCM overlaid in the same figure.

Transient Analysis `tran`: time = (0 s -> 3 us)

Name	Vis	VREFF
/VINN		1.73
/VINP		1.73
/VOCM		1.73
/VOUTN		1.73
/VOUTP		1.73



Do you notice any differential/CM ringing? Are both loops stable with adequate PM?

there is no ringing in differential output signal because phase margin is higher than 88.83° , so it will be an overdamped system.

Yes, both loops are stable with adequate PM as we saw in the STB analysis

Calculate the 1% settling time and compare it to the required specification. If the specification is not satisfied, what design changes could be a possible solution?

Test	Output	Nominal
FoldedCascode:closed:1	riseTime(v("/VODIFF" ?result ...	98.23n

required specification , so the specification is satisfied.

If the settling requirement is not met, you can increase the OTA gm (by raising tail current or device width) to boost GBW and speed up the response.

If ringing appears, add a small series damping resistor with the feedback capacitor or adjust the compensation network.

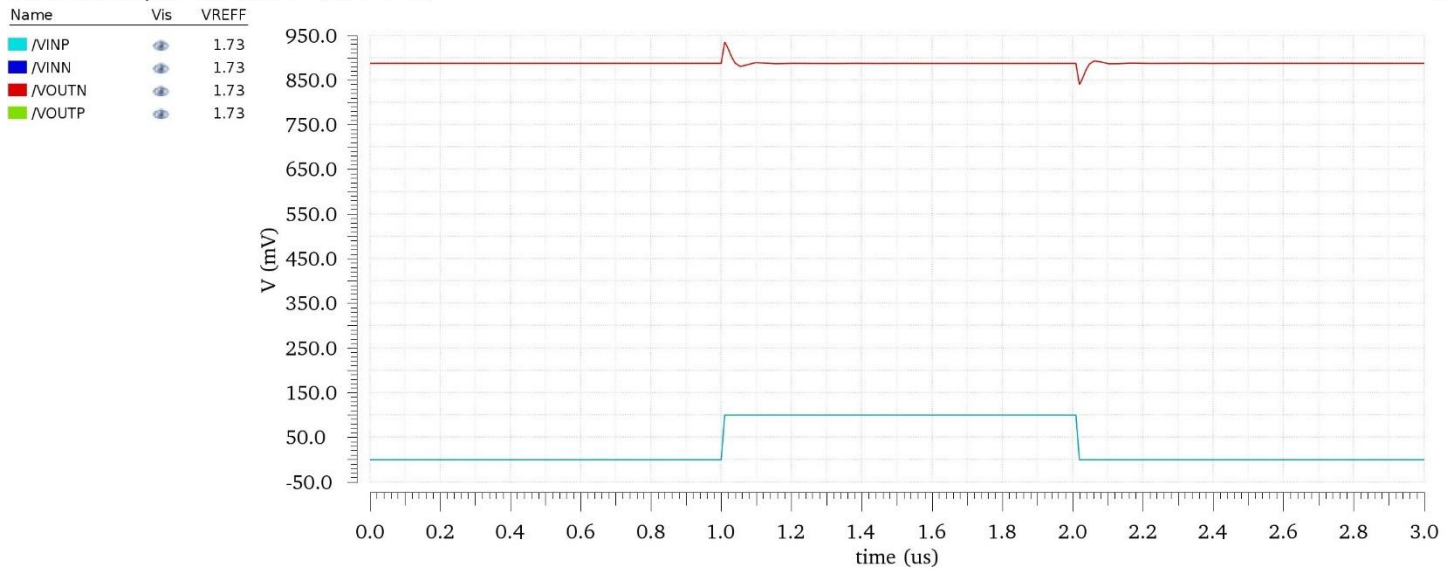
Reducing the effective output capacitance or adding a stronger output buffer can also help improve settling.

Differential and CMFB loops stability (transient analysis): CM input pulse

Plot the transient signals at VINP, VINN, VOUTP, VOUTN, and VOCM overlaid in the same figure.

Transient Analysis `tran`: time = (0 s -> 3 us)

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Do you notice any differential/CM ringing? Are both loops stable with adequate PM?

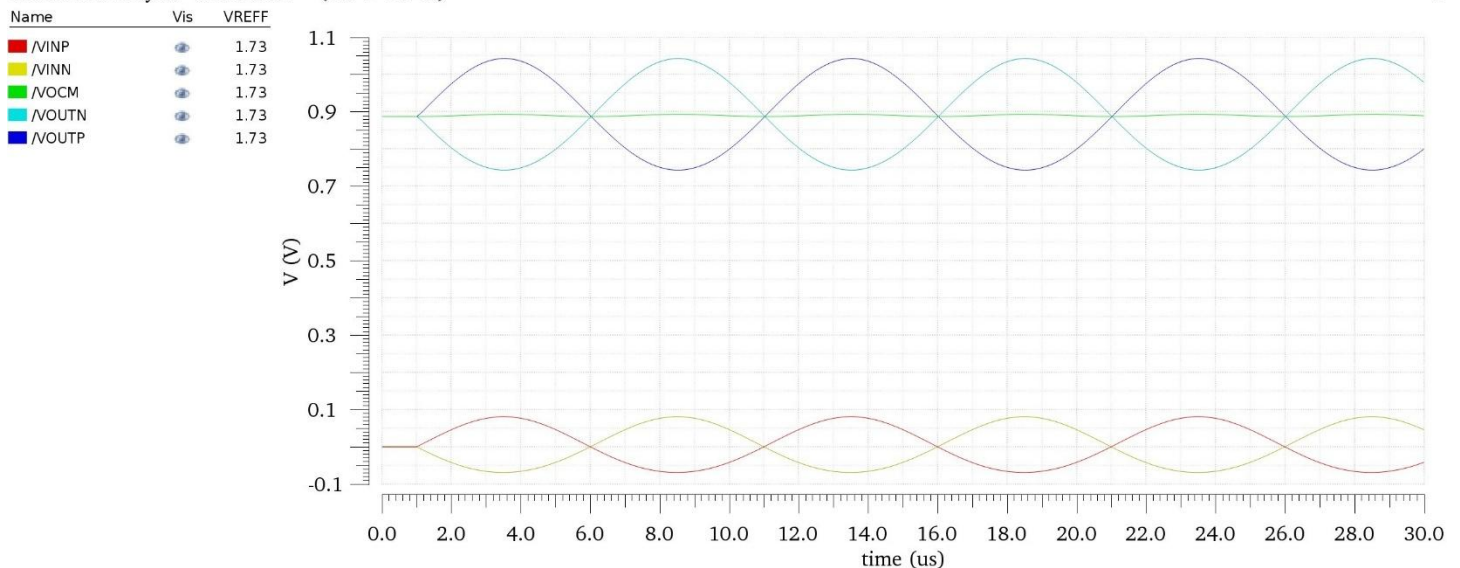
Yes, there is a very small ringing in CM output signal because its phase margin is lower than 57.39° . so it will be an under damped system.

3) Output swing:

Plot the transient signals at VINP, VINN, VOUTP, VOUTN, and VOCM overlaid in the same figure.

Transient Analysis `tran`: time = (0 s -> 30 us)

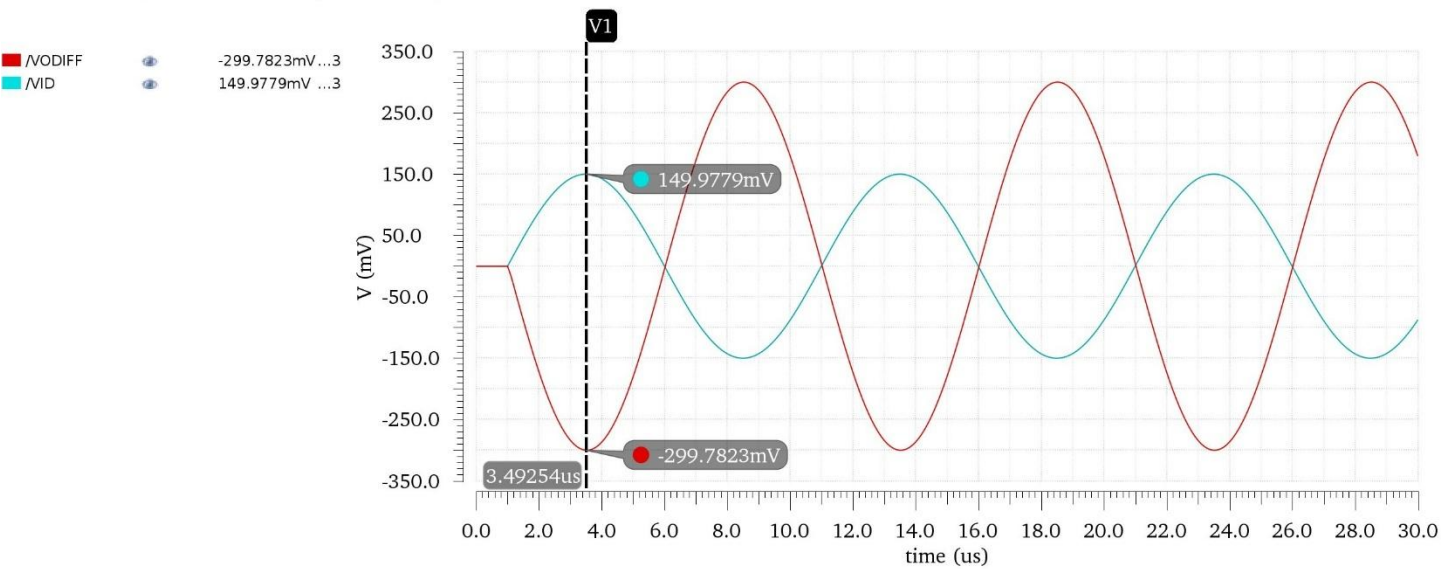
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Plot the transient signals at VIDIFF and VODIFF overlaid in the same figure.

Transient Analysis `tran`: time = (0 s -> 30 us)

1



Calculate the diff input and output peak-to-peak swings and the closed loop gain.

Test	Output	Nominal
Fully-DifferentialFoldedCascodeOTA:TB:1	peakToPeak(v("/VIDIFF" ?resu...	300m
Fully-DifferentialFoldedCascodeOTA:TB:1	peakToPeak(v("/VODIFF" ?res...	599.8m